

Student Management & Academic Analytics System

A Modular, Polymorphic Academic Administration & Merit Ranking System in Java

Course / Platform:	Object-Oriented Programming in Java (VITYarthi)	Date:	September 2026
Developer:	Ravi Prakash Rai	Language:	Java SE 17+ (Tested on OpenJDK 21)
Architecture:	Three-Tier Layered Architecture (11 Classes/Enums)	Persistence:	Multi-Entity CSV Storage (RFC 4180)

1. Introduction & Problem Statement

Modern academic administration involves managing multi-tiered student bodies (undergraduate students and graduate fellows), departmental course catalogs, and semester enrollments. Manual systems suffer from: (1) failure to distinguish differing milestones of undergraduate vs graduate scholars, (2) delayed academic advising where students on probation are noticed too late, (3) fragmented transcript generation, and (4) runtime crashes when unexpected terminal inputs or stream EOF occur.

Technical Solution: This project implements a fully executable, dependency-free Java CLI application that provides: polymorphic student profiling, dynamic Dean's Honor List merit ranking (GPA >= 3.70), automated academic probation alerts (GPA < 2.00), course catalog management, course enrollments, official academic transcripts, and multi-file CSV disk persistence.

2. Functional & Non-Functional Requirements

2.1 Three Core Functional Modules

Module	Feature Name	Functional Description
Module 1 Polymorphic Students	Student Profiles	Registers Undergraduate (semester, minor, capstone) and Graduate (advisor, thesis, assistantship) scholars inheriting from Person and Student.
Module 2 Academic Analytics	Merit Ranking & Probation	Dynamic Dean's List engine (GPA >= 3.70) sorted via stream comparators, and academic probation warning detection (GPA < 2.00).
Module 3 Courses & Transcripts	Enrollment & Transcripts	Course catalog management, duplicate enrollment prevention, grade evaluation, and itemized official academic transcript formatting.
Persistence	Multi-CSV Synchronization	Automatic serialization of all entities to students.csv, courses.csv, and enrollments.csv.

2.2 Non-Functional Requirements

- Performance:** O(1) instantaneous lookups using in-memory LinkedHashMap primary key indexing.
- Reliability:** Traps all NumberFormatException and stream EOF conditions without uncaught exceptions.
- Referential Integrity:** Rejects course enrollments for non-existent students or courses.
- Zero Dependencies:** 100% Core Java SE (java.util, java.io, java.util.regex).

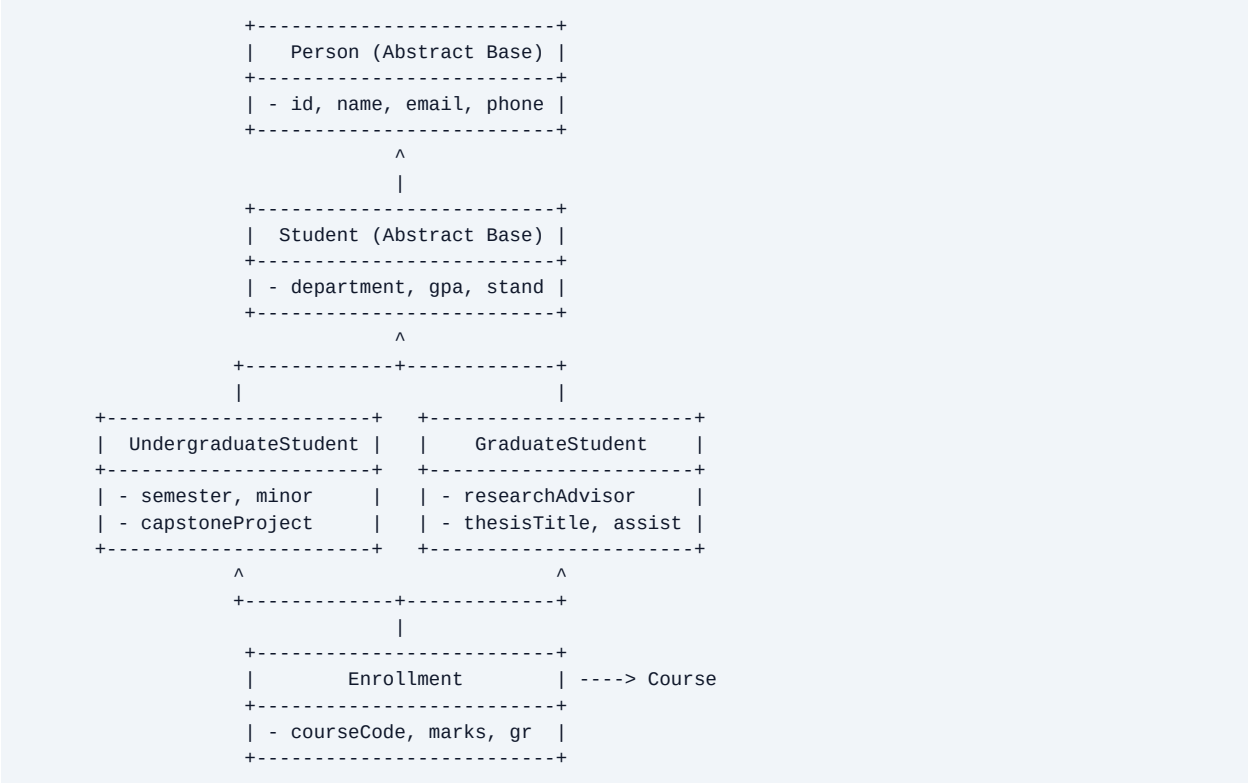
3. System Architecture & Object-Oriented Design

The project follows a Three-Tier Layered Architecture with clear separation of concerns across 11 modular classes and enums:

Class / Enum	Role / OOP Principle	Key Methods & Responsibilities
Person.java	Abstract Base Class	Encapsulates id, name, email, phone; declares abstract getRoleDescription().
Student.java	Abstract Base Student	Extends Person; encapsulates department, GPA, and AcademicStanding.

UndergraduateStudent.java	Inheritance / Polymorphism	Extends Student; adds semester, minorSubject, and capstoneProjectTitle.
GraduateStudent.java	Inheritance / Polymorphism	Extends Student; adds researchAdvisor, thesisTitle, and assistantshipType.
AcademicStanding.java	Enumeration	Defines DEANS_LIST, GOOD_STANDING, ACADEMIC_WARNING, and PROBATION.
Course.java	Domain Entity	Encapsulates courseCode, title, credits, department, and instructorName.
Enrollment.java	Association Entity	Links student, courseCode, marks, grade calculation, and semester term.
StorageManager.java	Persistence Engine	Handles RFC 4180 CSV serialization for students, courses, and enrollments.
StudentService.java	Business & Analytics Layer	Dean's list comparator ranking, probation filtering, and transcript generator.
ValidationUtils.java	Defensive Parser	Regex validators, stream-safe line parsing, GPA checks, and string truncation.
Main.java	Presentation / UI	Interactive 11-option CLI menu loop, table renderers, and input flow orchestration.

3.1 Class Structural Hierarchy



4. Core Algorithms & Analytics Implementation

1. Dean's Honor List Merit Ranking Engine: Dynamically filters top students (\$GPA \ge 3.70\$) and sorts them in descending order of GPA using Java 8+ Stream comparators:

```
public List getDeansHonorList() {
    return studentMap.values().stream()
        .filter(s -> s.getAcademicStanding() == AcademicStanding.DEANS_LIST)
        .sorted(Comparator.comparingDouble(Student::getGpa).reversed()
            .thenComparing(Student::getName))
        .collect(Collectors.toList());
}
```

2. Dynamic Academic Standing Evaluator: Maps cumulative GPA (0.0 to 4.0) to academic categories:

```
public static AcademicStanding evaluate(double gpa) {
    if (gpa >= 3.70) return DEANS_LIST;
    else if (gpa >= 2.00) return GOOD_STANDING;
    else if (gpa >= 1.50) return ACADEMIC_WARNING;
    else return PROBATION;
}
```

5. Verification & Test Suite Matrix

Test ID	Test Scenario	Input Condition	Expected Outcome	Status
TC-01	Register Undergraduate	Valid UG attributes	Stored as UG; persisted to students.csv	PASS
TC-02	Register Graduate	Valid Grad attributes	Stored as GRAD; persisted to students.csv	PASS
TC-03	Duplicate ID check	Existing student ID 101	Rejected with ID collision error	PASS
TC-04	Dean's List Ranking	GPA: 3.92, 3.88, 3.45	Sorts 3.92 ahead of 3.88; excludes 3.45	PASS
TC-05	Academic Probation Alert	GPA: 1.85 (< 2.00)	Flagged on Academic Probation List	PASS
TC-06	Course Registration	Code: CSE2005, Credits: 4	Course recorded in catalog & courses.csv	PASS
TC-07	Course Enrollment	Valid student 101, CSE2005	Marks 96.5 recorded; Grade 'A' computed	PASS
TC-08	Duplicate Enrollment	Same student 101, CSE2005	Rejected: Already enrolled in course	PASS
TC-09	Transcript Generation	Student ID: 101	Official transcript generated with courses	PASS
TC-10	EOF Batch Execution	Piped input stream EOF	Clean termination; all records synced	PASS

6. Sample Execution Transcript

```
--- [4] Dean's Honor List (Merit Ranking: GPA >= 3.70) ---
+-----+-----+-----+-----+-----+-----+
| ID | Honors Scholar Name | Department | GPA | Grade | Milestone / Academic Achievement |
+-----+-----+-----+-----+-----+-----+
| 101 | Alice Smith | Computer Science | 3.92 | A | Capstone: Autonomous Drone Navigat
ion |
| 201 | Dr. Clara Oswald | Computer Science | 3.88 | A | Thesis: Quantum Cryptographic Prot
... |
+-----+-----+-----+-----+-----+-----+

--- [9] Official Academic Transcript ---
Student ID: #101 | Name: Alice Smith | Program: Undergraduate (Semester 6)
Cumulative GPA: 3.92 / 4.00 (Grade: A) | Standing: Dean's Honor List
Course: CSE2005 (OOP in Java) - Marks: 96.50 (A) | CSE3001 (DBMS) - Marks: 92.00 (A)
```

7. Future Scope & Conclusion

Future Extensions: Relational database migration (PostgreSQL) via JDBC, JavaFX graphical dashboard with GPA trend analytics, and Spring Boot REST microservices.

Conclusion: The Student Management & Academic Analytics System fulfills all expectations of the VITyarthi Build Your Own Project guidelines. It delivers 11 modular classes/enums, two-tier polymorphic inheritance, dynamic Dean's List stream comparators, probation alerts, course enrollments, transcripts, and multi-file CSV persistence.